16.07 Dynamics Problem Set 3 Solutions

Problem 1: Mass Properties of a T-Shaped Body

(a) Mass

$$m = m_A + m_B = 588.8 + 384.0 = \boxed{972.8 \text{ g}}$$

(b) Center of mass

Measure positions along x from the junction point Q, at the center of the face where the two boxes meet. Because each box has uniform density, its CoM is at its geometric center: C_A is at $x = -4$ cm (half of box A's 8 cm width) and C_B is at $x = 7.5$ cm (half of box B's 15 cm length). By symmetry, all three centers of mass lie on the x axis with $y = z = 0$. The CoM of the T-shaped body is the mass-weighted average:

$$x_O = \frac{m_A x_A + m_B x_B}{m} = \frac{588.8(-4) + 384.0(7.5)}{972.8} = \frac{524.8}{972.8} = 0.5395 \text{ cm}.$$

So O lies 0.54 cm from the junction, just inside the stem. The position vectors from O to the box centers of mass are

$$\boxed{\mathbf{r}_1 = \begin{bmatrix} -4.539 \\ 0 \\ 0 \end{bmatrix} \text{ cm}}, \qquad \boxed{\mathbf{r}_2 = \begin{bmatrix} 6.961 \\ 0 \\ 0 \end{bmatrix} \text{ cm}}$$

(c) Inertia matrix

The parallel axis theorem shifts each box's inertia matrix from its own CoM to O:

$$\mathbf{J} = \sum_i \left[\mathbf{J}_i - m_i S(\mathbf{r}_i)^2 \right].$$

The shifted matrices of the two boxes about O are

$$\mathbf{J}_A^{(O)} = \begin{bmatrix} 27{,}180 & 0 & 0 \\ 0 & 4{,}367 + 12{,}133 & 0 \\ 0 & 0 & 29{,}100 + 12{,}133 \end{bmatrix},$$

$$\mathbf{J}_B^{(O)} = \begin{bmatrix} 2{,}848 & 0 & 0 \\ 0 & 8{,}000 + 18{,}604 & 0 \\ 0 & 0 & 9{,}248 + 18{,}604 \end{bmatrix},$$

in g cm^2. Adding them,

$$\boxed{\mathbf{J} = \begin{bmatrix} 30{,}028 & 0 & 0 \\ 0 & 43{,}105 & 0 \\ 0 & 0 & 69{,}086 \end{bmatrix} \text{ g cm}^2 \approx \begin{bmatrix} 3.003 & 0 & 0 \\ 0 & 4.310 & 0 \\ 0 & 0 & 6.909 \end{bmatrix} \times 10^4 \text{ g cm}^2}$$

Problem 2: The Bifilar Pendulum

Equations of Motion

(a) Geometry and energies

In the top view, the two ends of a string and the axis form an isosceles triangle with two sides of length d and an included angle θ. The third side is the horizontal displacement s, so by the law of cosines

$$s^2 = d^2 + d^2 - 2d^2\cos\theta = 2d^2(1-\cos\theta), \qquad \text{equivalently} \qquad s = 2d\sin(\theta/2).$$

Each string has fixed length ℓ and horizontal extent s, so its vertical extent is $\sqrt{\ell^2 - s^2}$. At rest the vertical extent is ℓ, so the body rises by

$$\boxed{h(\theta) = \ell - \sqrt{\ell^2 - 2d^2(1-\cos\theta)}}$$

Taking the rest position as the reference, the potential energy is $U = mgh(\theta)$. The body rotates about the vertical axis at rate $\dot\theta$ and translates vertically at rate

$$\dot h = \frac{dh}{d\theta}\,\dot\theta = \frac{d^2\sin\theta}{\sqrt{\ell^2 - 2d^2(1-\cos\theta)}}\,\dot\theta,$$

so the kinetic energy is

$$T = \frac{1}{2}J\dot\theta^2 + \frac{1}{2}m\dot h^2 = \frac{1}{2}\left[J + \frac{md^4\sin^2\theta}{\ell^2 - 2d^2(1-\cos\theta)}\right]\dot\theta^2.$$

The exact Lagrangian is

$$\boxed{\mathcal{L}(\theta,\dot\theta) = \frac{1}{2}\left[J + \frac{md^4\sin^2\theta}{\ell^2 - 2d^2(1-\cos\theta)}\right]\dot\theta^2 - mg\left(\ell - \sqrt{\ell^2 - 2d^2(1-\cos\theta)}\right)}$$

(b) Small-angle expansion

Expanding $1-\cos\theta = \frac{1}{2}\theta^2 - \frac{1}{24}\theta^4 + \cdots$ gives $s^2 = d^2\theta^2 + \mathcal{O}(\theta^4)$, and then

$$\sqrt{\ell^2 - s^2} = \ell\sqrt{1 - \frac{s^2}{\ell^2}} = \ell - \frac{s^2}{2\ell} + \mathcal{O}(s^4), \qquad h \approx \frac{d^2\theta^2}{2\ell}, \qquad U \approx \frac{mgd^2}{2\ell}\,\theta^2.$$

In the kinetic energy, the rotational term $\frac{1}{2}J\dot\theta^2$ is already quadratic. The vertical-translation term is

$$\frac{1}{2}m\dot h^2 \approx \frac{1}{2}\,\frac{md^4}{\ell^2}\,\theta^2\dot\theta^2,$$

which is *fourth* order in the small quantities, so it drops out. Physically, the body's height changes only at second order in θ, so its vertical velocity is itself second order ($\dot h \propto \theta\dot\theta$), and the kinetic energy of that motion is fourth order. The surviving quadratic Lagrangian is

$$\boxed{\mathcal{L}_2 = \frac{1}{2}J\dot\theta^2 - \frac{mgd^2}{2\ell}\,\theta^2}$$

(c) Linearized equation of motion

With $\partial\mathcal{L}_2/\partial\dot{\theta} = J\dot{\theta}$ and $\partial\mathcal{L}_2/\partial\theta = -mgd^2\theta/\ell$, the Euler–Lagrange equation

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}_2}{\partial\dot{\theta}}\right) - \frac{\partial\mathcal{L}_2}{\partial\theta} = 0$$

gives

$$J\ddot{\theta} + \frac{mgd^2}{\ell}\,\theta = 0 \qquad \Longrightarrow \qquad \boxed{\ddot{\theta} + \frac{mgd^2}{J\ell}\,\theta = 0}$$

(d) Natural frequency and period

Reading off the coefficient of θ in the equation of motion,

$$\boxed{\omega = d\sqrt{\frac{mg}{J\ell}}}, \qquad \boxed{P = \frac{2\pi}{\omega} = \frac{2\pi}{d}\sqrt{\frac{J\ell}{mg}}}$$

For the T-shaped body, with $m = 972.8$ g and $J = J_{zz} = 69{,}086\ \mathrm{g\,cm^2}$ from Problem 1, $g = 981\ \mathrm{cm/s^2}$, $\ell = 200$ cm, and $d = 7.5$ cm:

$$\omega = 7.5\sqrt{\frac{972.8 \times 981}{69{,}086 \times 200}} = \boxed{1.971\ \mathrm{rad/s}}, \qquad P = \frac{2\pi}{\omega} = \boxed{3.188\ \mathrm{s}}$$

(e) Measuring the inertia

Squaring the expression for P from part (d) and solving for J:

$$P^2 = \frac{4\pi^2}{d^2}\,\frac{J\ell}{mg} \qquad \Longrightarrow \qquad \boxed{J = \frac{mgd^2P^2}{4\pi^2\ell}}$$